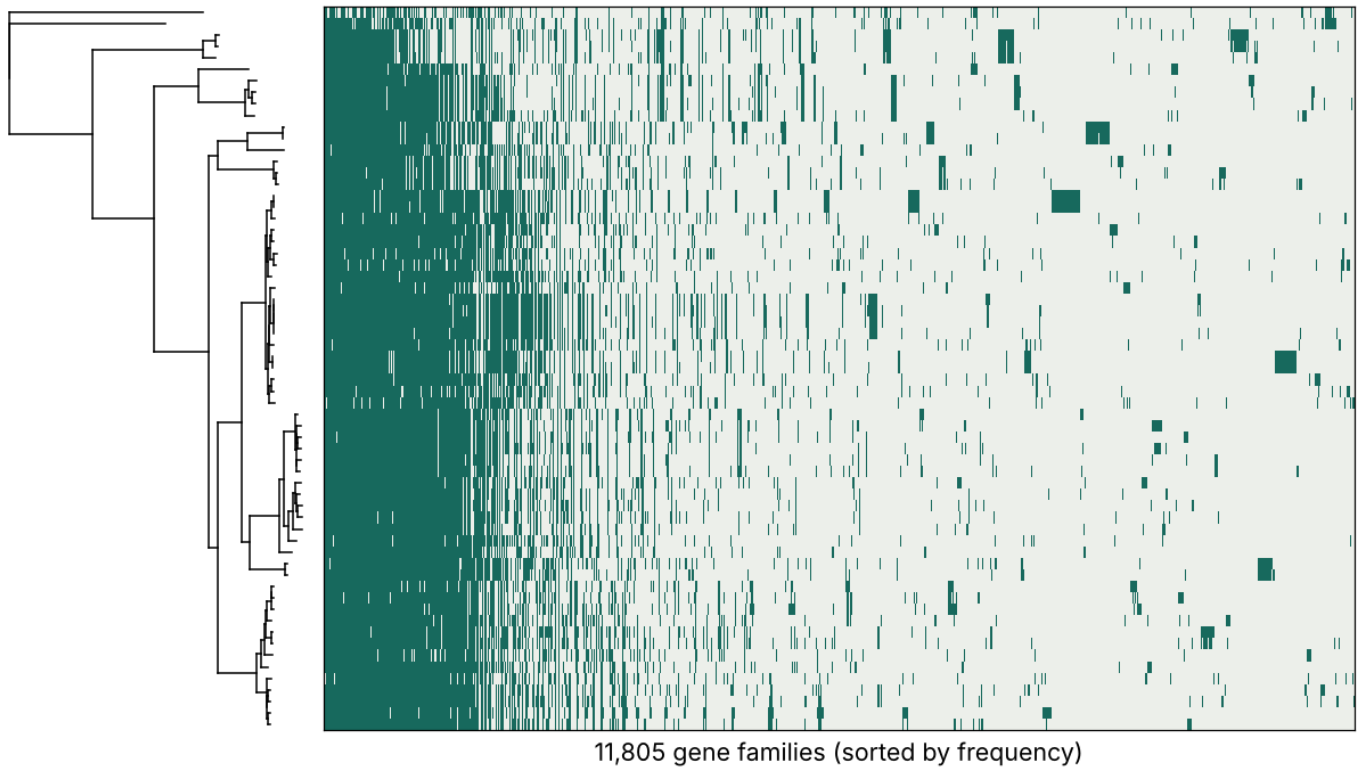
TRR356 WORKSHOP · PART 2

Bacterial phylogenomics

Part 2: Gain and loss, co-evolution and gene-content GWAS

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Our gene-content variation, written as phyletic patterns



From Part 1: a matrix

63 genomes × 26,703 families; the 11,805 variable ones (2–62 genomes) are shown, ordered by the Microbializer tree.

Each column is a character

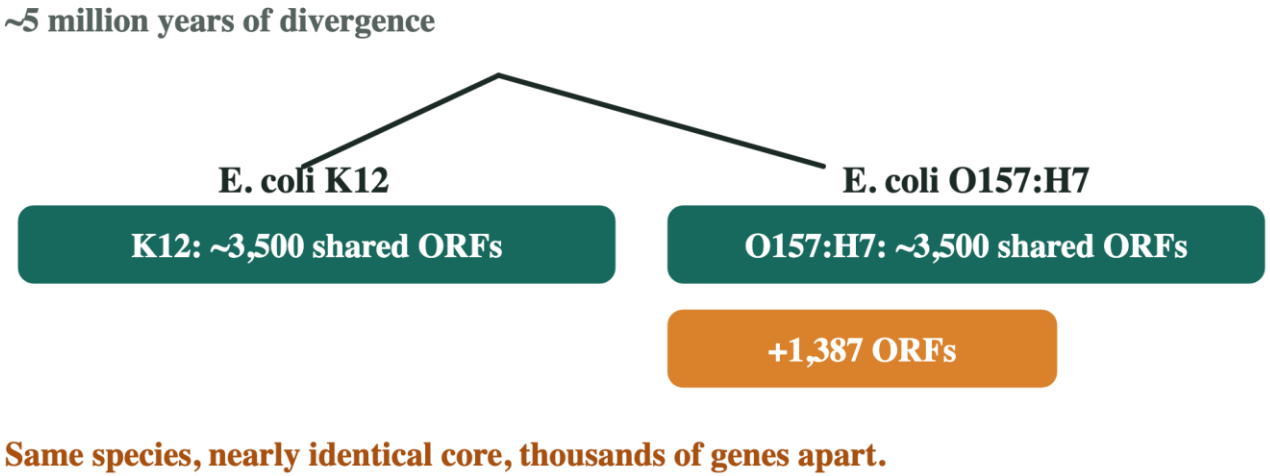
A gene family is a binary trait, 0 or 1, evolving along the tree: the **phyletic pattern**.

Blocks and patches

Blocks shared by a clade: vertical inheritance. Scattered patches: gains, losses, transfer.

Real Pseudomonas run (author's Microbializer output); columns sorted by frequency, rows in tree order

Why gene gain and loss? Lateral transfer shapes genomes



Gene content evolves fast

Two *E. coli* strains, ~5 million years apart, share ~3,500 ORFs; O157:H7 carries **1,387 more**.

Mostly lateral gene transfer

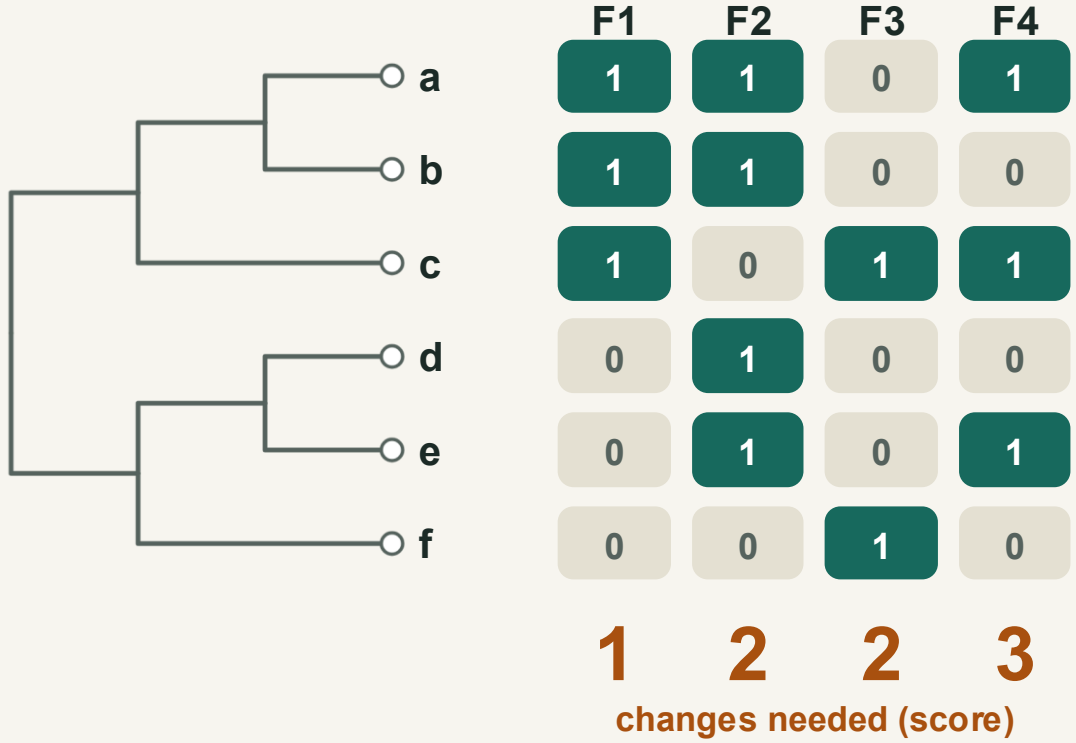
Many of the extra genes were acquired horizontally: methods to detect this look for atypical composition, or follow gene presence along a tree.

This is the gene-content signal that ANI hides (Part 1). To study it, write each family as **0/1 along the tree** (phyletic pattern) and model gain and loss.

Perna et al. (2001) Nature 409:529; slide ideas: Tal Pupko, lecture “Comparative genomics analyses provide novel insights on lateral gene transfer”

Scoring a gene family on a fixed tree

Fixed phylogenetic tree (an input: it is not changed or estimated here)



Maximum parsimony

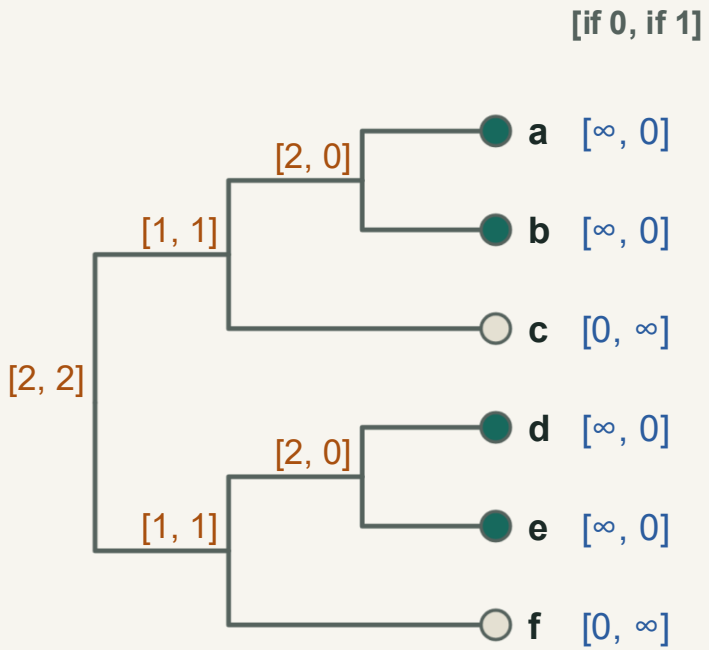
Explain the pattern with the fewest possible changes.

The best explanation minimizes **homoplasy**: convergent or parallel evolution (the same state arising independently) and reversals. When building a tree, the best tree is the one that needs the fewest evolutionary steps for all the data.

Here the tree is given. The **score** of a family is the smallest number of changes (gains 0→1, losses 1→0) that explains its tips. Score 1: inherited together. Score 2 or more: homoplasy.

Toy example, unit costs. Parsimony scoring on a fixed tree: Fitch (1971) Syst Zool 20:406–416; Sankoff (1975) SIAM J Appl Math 28:35–42.

How the score is computed: Sankoff's algorithm



How to read [x, y]: the fewest changes needed **below that node** if the node is **absent (x)** or **present (y)**. ∞ = impossible.

Cost matrix: gain 0→1 = 1, loss 1→0 = 1, no change = 0.
Rule: for each child take the cheaper of: it keeps the node's state (+0) or it differs (+cost of the change), then add the children.

1 · Tips. Tip a is present: **[∞, 0]** = it cannot be absent (∞) and needs 0 changes if present.

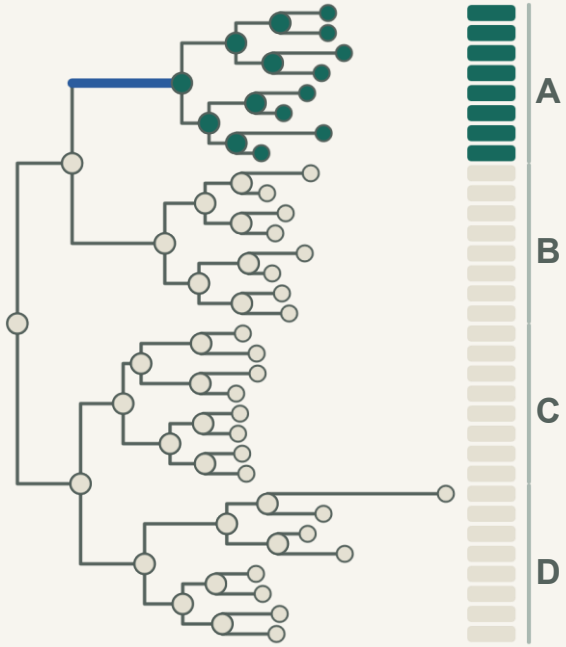
2 · Up the tree. Node (a, b) if **absent**: a must change (1) + b must change (1) = **2**. If **present**: 0 + 0 = **0**. So **[2, 0]**. Repeat for every internal node.

3 · Root. [2, 2]: the smaller number is the parsimony score, **2** = two gains.

Different costs for gains and losses? Change the matrix and run the same algorithm again. That is what the next slides vary.

Sankoff (1975) SIAM J Appl Math 28:35. Click to step. Unit costs; family F2 of the previous slide.

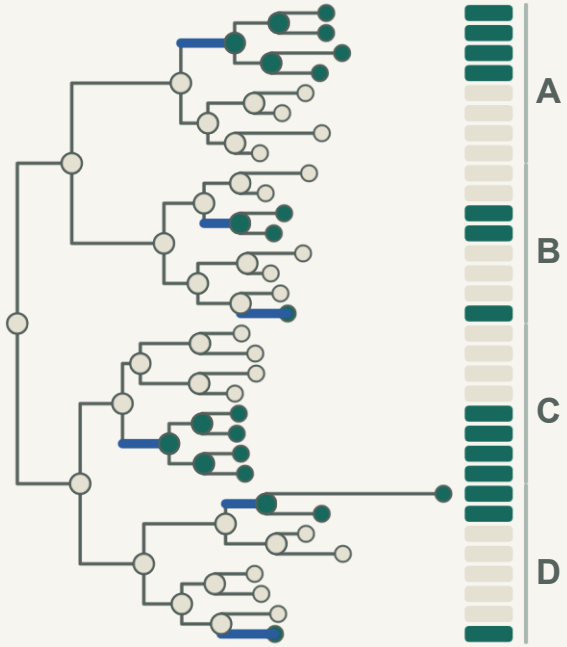
Fitch: Sankoff with unit costs, step by step



Family 10

present in 8 of 32 genomes

1 gain



Family 0

present in 14 of 32 genomes

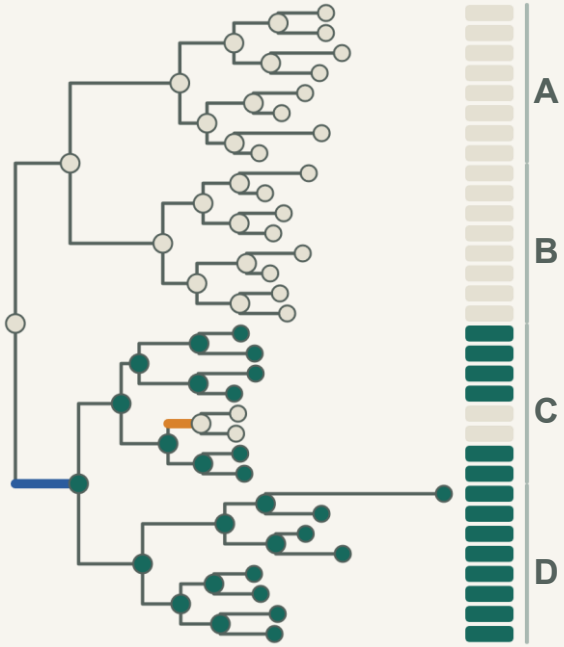
6 gains

gain
present
absent

3 · Mark the changes. Family 10: **one gain** on the branch to clade A. Family 0: **six independent gains**, although 14 genomes carry it: **homoplasy**.

Simulated data (toy dataset, 32 genomes, 60 families); Fitch (1971) Syst Zool 20:406; Sankoff (1975) SIAM J Appl Math 28:35. Click to step.

Parsimony needs costs: gains versus losses



A · absent at the root

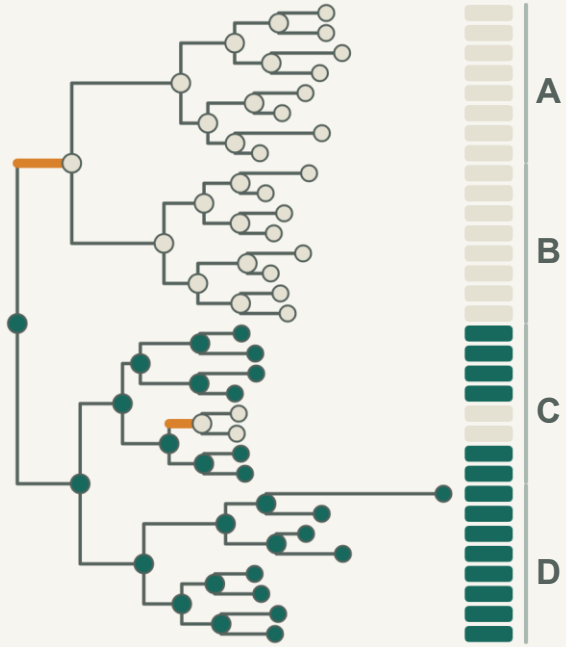
1 gain + 1 loss

Equal costs (1 and 1)

cost 2

Gain 3, loss 1

cost 4



B · present at the root

0 gains + 2 losses

Equal costs (1 and 1)

cost 2

Gain 3, loss 1

cost 2

preferred

gain

loss

Which one is chosen depends on the **costs**. If a gain is three times as costly as a loss, B wins. The costs are arbitrary: a probabilistic model **estimates gain and loss rates from the data**. That is GLOOME.

Simulated data (toy dataset); Sankoff parsimony with unit costs and with gain cost 3, loss cost 1. Click to change the costs.

The toy dataset (simulated)

32

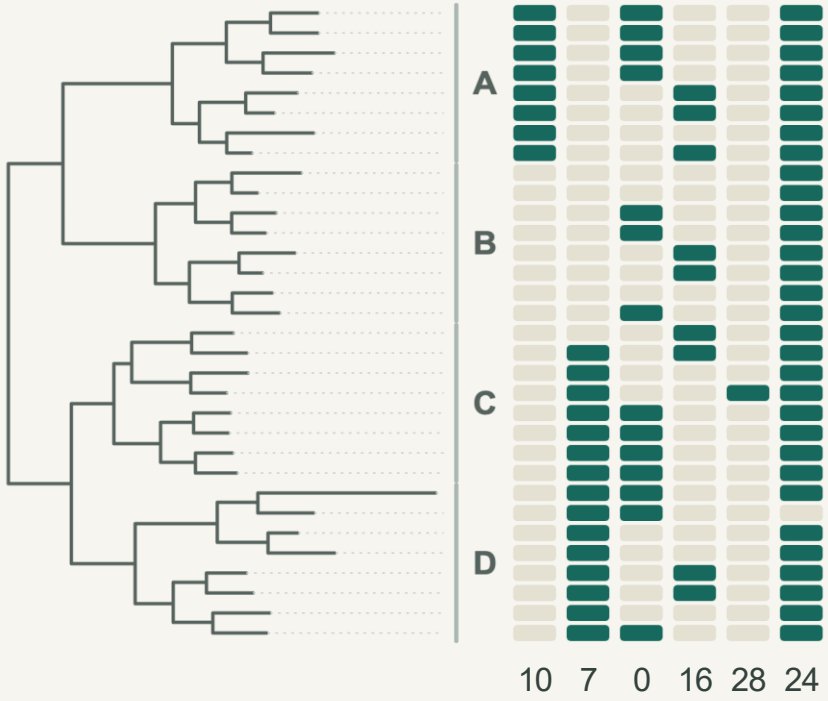
genomes in four clades (A to D)

60

gene families, each present (1) or absent (0) in every genome

1

rooted tree that tells us who is related to whom

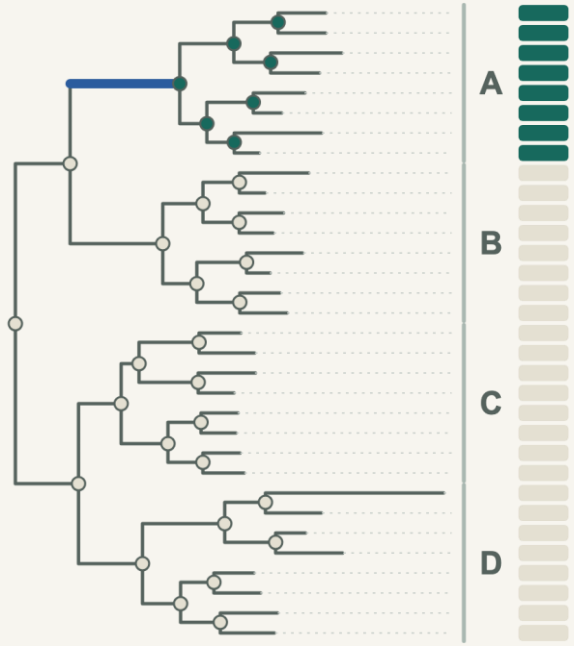


Not real bacteria: the families were simulated on this tree, so we know the true history. Annotation is invented.

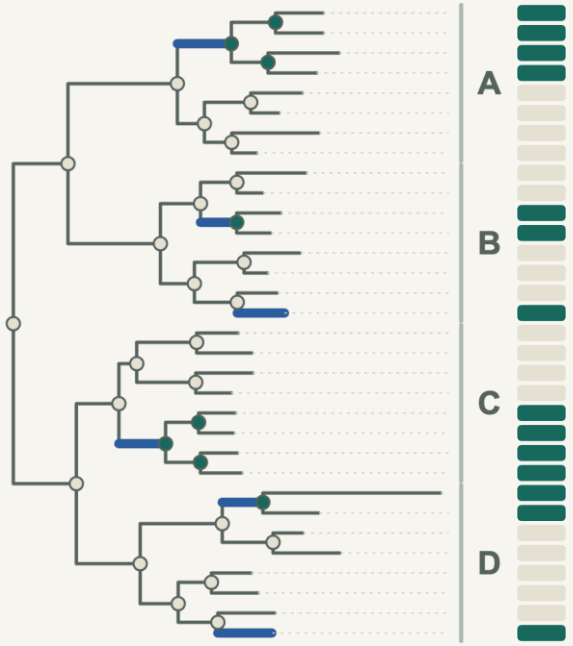
Source: simulated data run through GLOOME (Nodes.tsv, PhylogeneticTree.nwk); six of the 60 families shown, GLOOME numbering 0 to 59

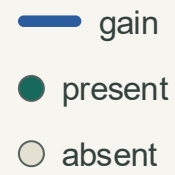
Two gene families, one tree

Family 10: present in clade A. One gain explains it.



Family 0: scattered. Six gains are needed.





Source: simulated data, GLOOME ancestral states in Nodes.tsv; steps in ParsimonyAndHomoplasyScores.tsv

Counting changes on a tree

M · minimum possible steps

A gene that is present in some genomes and absent in others needs at least one change, whatever the tree. So M = 1.

S · steps on this tree

The fewest gains and losses that explain the pattern on the tree we were given. This is the parsimony score.

Family 10: M = 1, S = 1. One gain on the branch to clade A.

Family 0: M = 1, S = 6. Six gains in six separate lineages.

Source: ParsimonyAndHomoplasyScores.tsv, families 10 and 0

Homoplasy: the extra changes

Maximum parsimony explains the pattern with the **fewest changes**. Each change beyond the minimum is **homoplasy**: convergence, parallel gains, reversals.

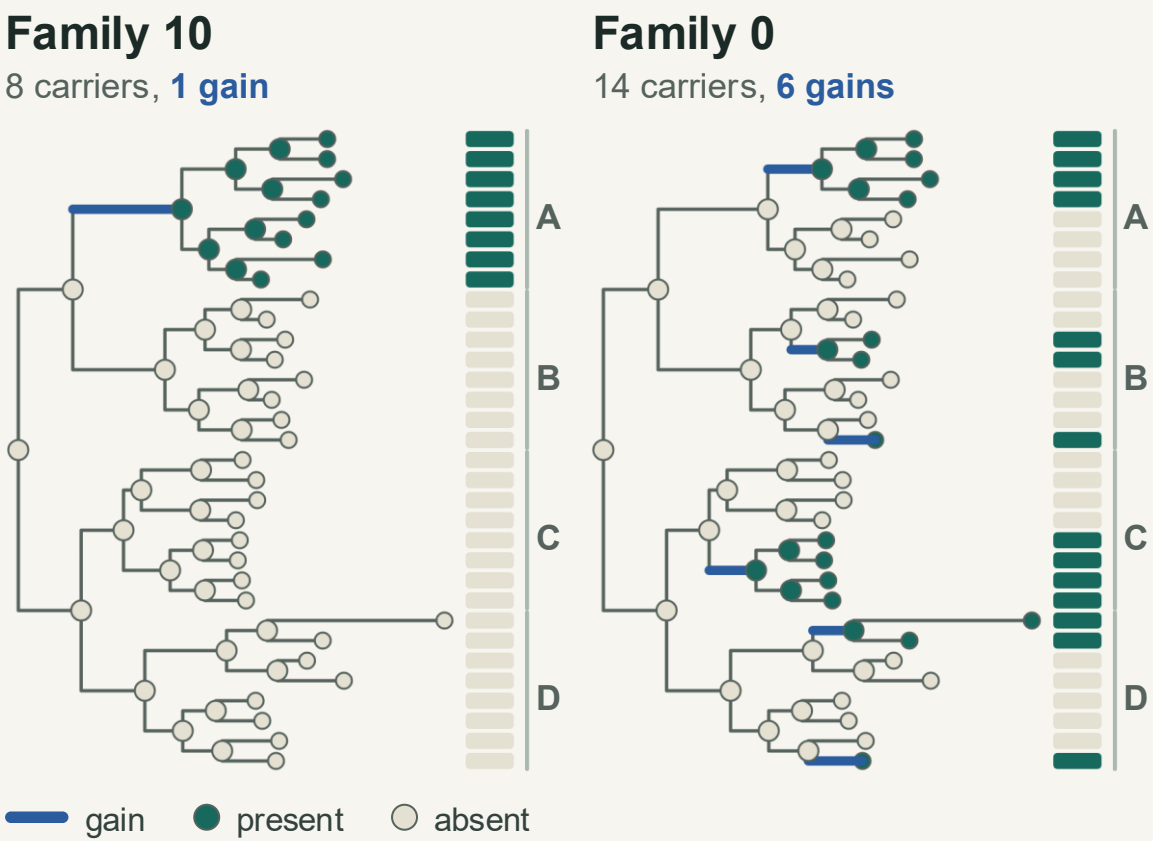
H = S − M
extra changes

0
Family 10
1 − 1

5
Family 0
6 − 1

Why would a gene appear more than once?

- Independent gains: transfer or convergence
- Repeated losses
- Tree does not match the gene's history
- Noise: annotation, assembly, split orthogroups

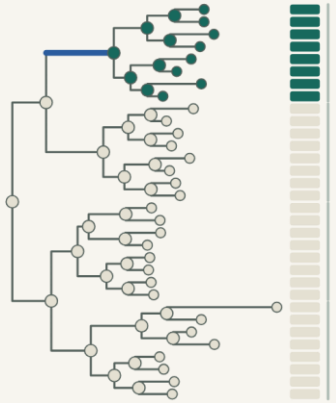


Simulated data (toy dataset), families 10 and 0. S = changes needed, M = minimum (1 for a family inherited as one block), H = S − M extra steps (Camin & Sokal 1965 Evolution 19:311–326; Kluge & Farris 1969 Syst Zool 18:1–32; Farris 1989 Cladistics 5:417–419).

Consistency index

CI = M / S minimum changes (M = 1) divided by changes needed on the tree (S)

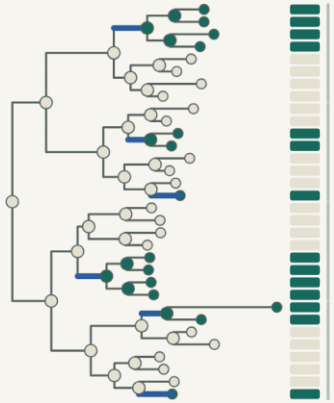
1.0 = the family follows the tree completely; lower values = a more scattered family.



Family 10

S = 1 (1 gain)

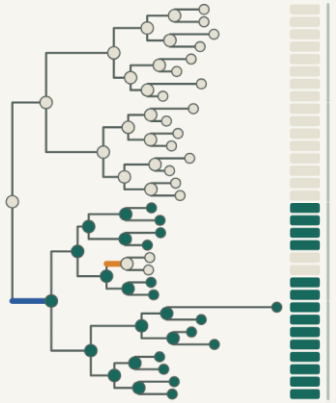
CI = 1.00



Family 0

S = 6 (6 gains)

CI = 0.17



Family 11

S = 2 (1 gain, 1 loss)

CI = 0.50

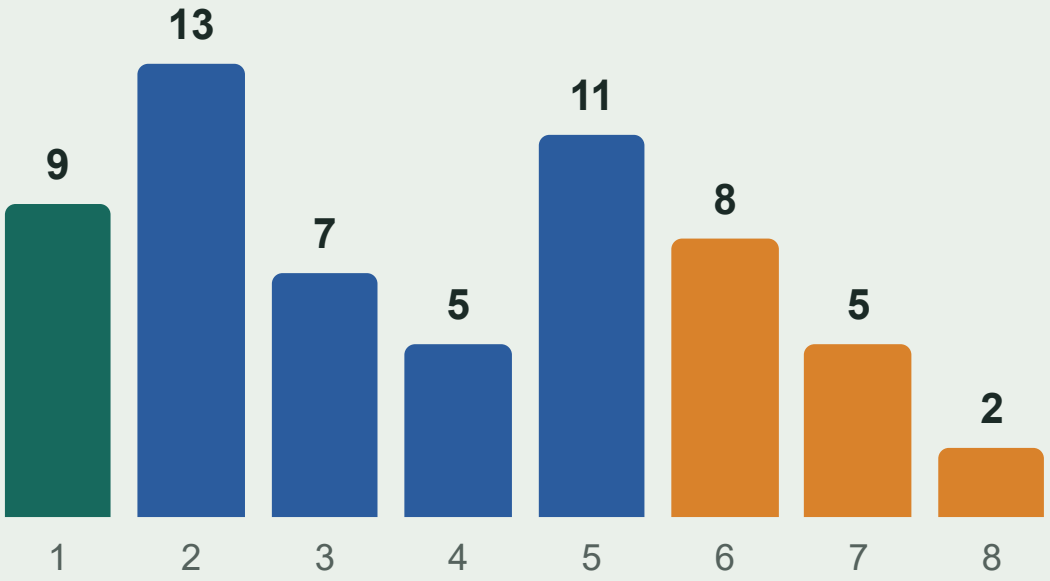
gain loss present absent

All 60 families together:

230	60	170	0.26
steps needed (S)	minimum possible (M)	extra steps (H)	consistency index (CI)

Simulated data (toy dataset); ParsimonyAndHomoplasyScores.tsv, TOTAL row. S, M, CI follow the standard s, m, CI: CI = m/s, Kluge & Farris (1969) Syst Zool 18:1–32; extra steps h = s − m, Camin & Sokal (1965) Evolution 19:311–326; Farris (1989) Cladistics 5:417–419.

Consistency across the 60 families



Horizontal axis: steps needed on the tree (S). Bars show how many families need that many.

Only 9 of 60 families fit the tree perfectly (CI 1.00).

Families 0, 32, 38, 40 and 58 need 6 to 8 steps. Remember them: they come back later.

Source: ParsimonyAndHomoplasyScores.tsv

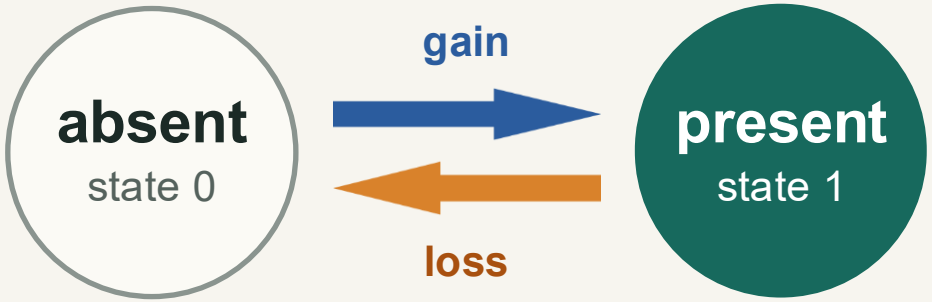
From counting to modelling

Counting (parsimony)	Modelling (GLOOME)
Every branch is equally likely to change	Longer branches have more time for change
One minimal solution is reported	Every possible history gets a probability
Gain and loss cost the same	Gain–loss balance is estimated (π1)
All gene families are treated alike	Each family gets its own rate

In the toy tree, D1’s branch is about 4 times longer than the typical tip branch (0.31 versus 0.08), yet parsimony counts it like any other branch.

Source: PhylogeneticTree.nwk, toy run

GLOOME's model in plain words



GLOOME estimates the probabilities of **0→1 changes (gain events)** and **1→0 changes (loss events)** by modelling evolution as a **continuous-time Markov process** along the tree.

A longer branch means more time, so a higher chance of a gain or a loss. The balance between gain and loss (π1) is estimated from the data.

π1 = 0.45

Equilibrium share of present genes: a bit below one half.

α = 2.96

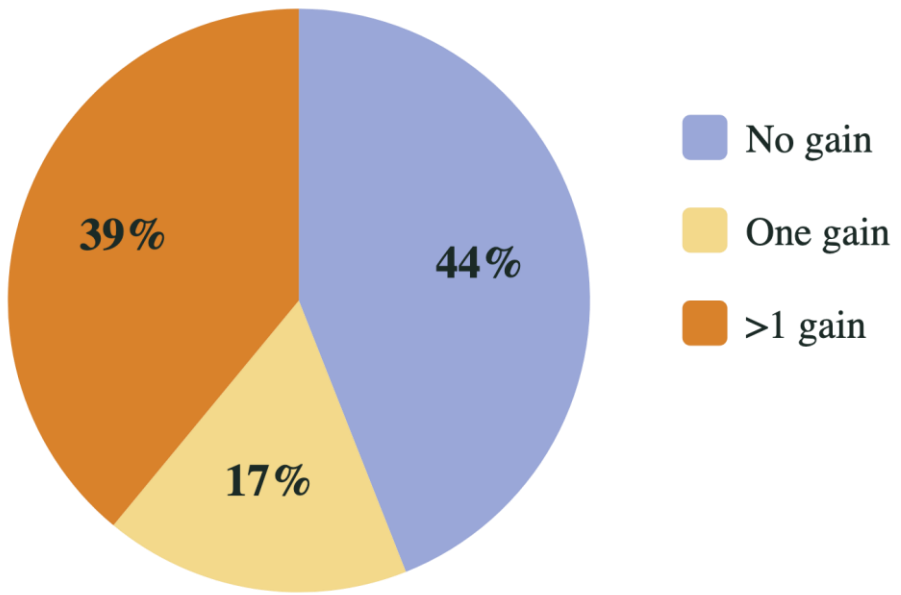
Shape of the gamma distribution of rates among families: four rate classes, from 0.39 to 1.80. A smaller α means more rate variation.

Branch lengths × 0.78

The tree was shrunk by about 22%: GLOOME rescales the tree to fit.

Source: TreeAttributes.tsv (π1, Γ shape α, number of rate categories, branch-length coefficient)

Prevalence of gene gain in microbial evolution



How many times was a family acquired?

4,873 gene families in 66 microbes. Stochastic mapping on the tree counts gain events per family.

Most families are not “once and done”

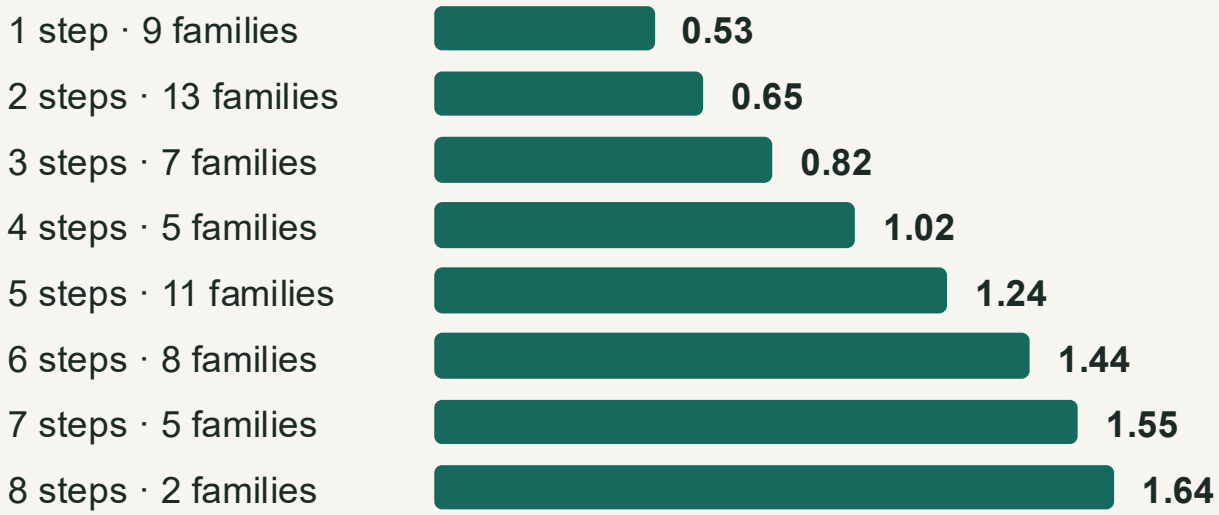
39% were gained more than once, **44%** never: lateral transfer is a major force, and rates differ among families.

This is why GLOOME gives each family its own rate (Gamma).

Cohen and Pupko (2010) Mol Biol Evol 27:703 (GLOOME, stochastic mapping); numbers from the author’s lecture

Rates of change per gene family

Average GLOOME rate, by number of steps



r = 0.98

between homoplasy (H) and GLOOME's rate across the 60 families

> Both views agree: families that need more steps are the ones GLOOME assigns higher rates.

Source: PosteriorRates.tsv and ParsimonyAndHomoplasyScores.tsv

What GLOOME returns

File	What it tells you
ParsimonyAndHomoplasyScores.tsv	M, S, H and CI for each family
PosteriorRates.tsv	Speed of each family
TreeAttributes.tsv and LogLikelihood.tsv	Fitted parameters and model fit
Nodes.tsv	Most likely state at every node
Branches.tsv and BranchPositionProbabilities.tsv	Gain and loss probability per branch
Coevolution.tsv	Pairs of families that co-evolve

In the toy run: 63 nodes (32 tips), 62 branches, 60 families and 1,770 family pairs tested for co-evolution.

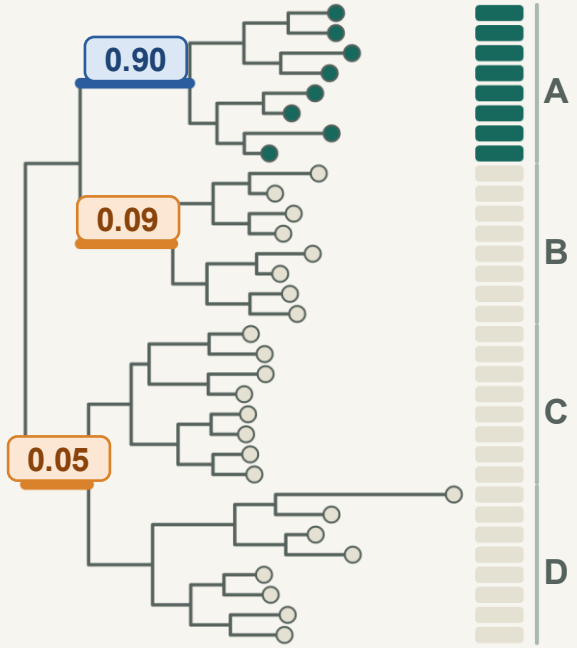
Source: GLOOME run 17936473625665, downloaded result files

Events as probabilities

gain loss present absent

Family 10 · S = 1

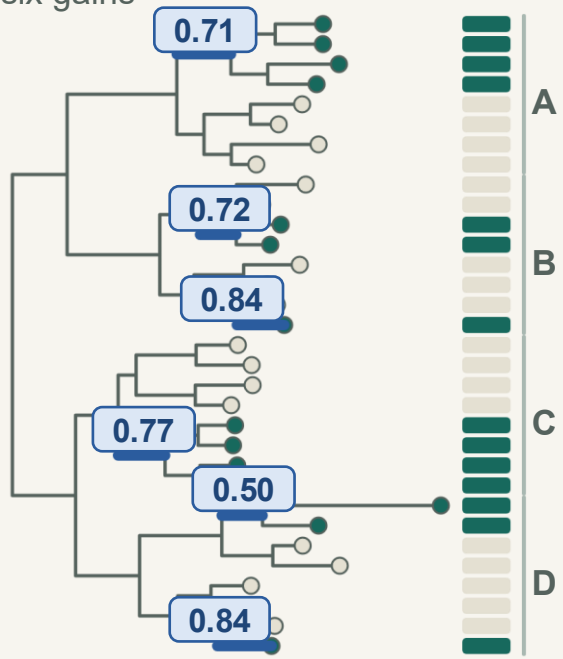
tree-like



One clear gain on the branch to clade A. The losses elsewhere are unlikely.

Family 0 · S = 6

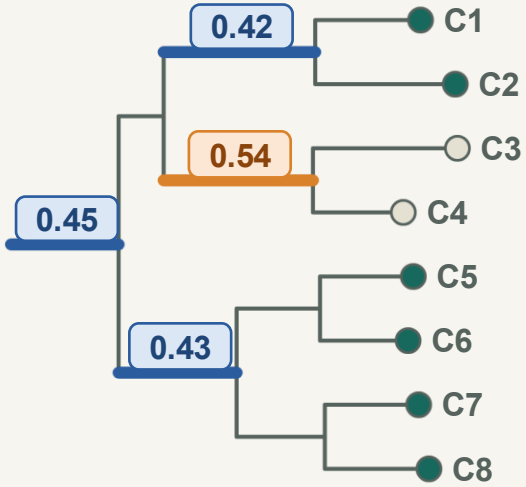
six gains



Six gains, none certain. The gains in single genomes (B8, D8) are the clearest.

Family 5 · S = 2

where was the gain? **Zoom on clade C only (8 of 32 genomes)**



Two stories fit: one gain in clade C and a loss in C3 and C4, or separate gains in C1 and C2 and in C5 to C8.

Number = probability of that event on that branch. Probabilities on different branches are alternatives, not shares of 100%.

Source: GLOOME BranchPositionProbabilities.tsv (toy run), top branch events per family, drawn on the toy tree

From single families to pairs

So far: one family at a time

Where did this family gain or lose? How many times? How certain are we?

Now: two positions together

Do two families gain and lose on the same branches, more often than chance?

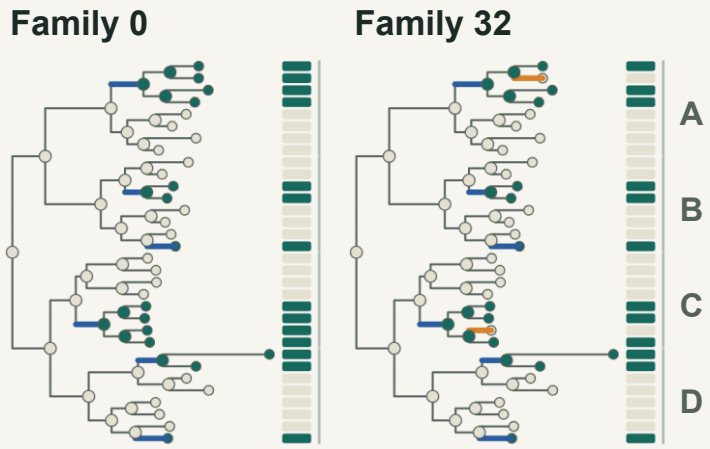
What moves together, and why

- Genes of one operon, or neighbours in the genome (synteny), gained and lost as a block
- Members of one pathway or complex that only work together
- Mobile elements and genomic islands that carry several genes at once

Next: a toy dataset with planted modules, to see what the method can and cannot recover

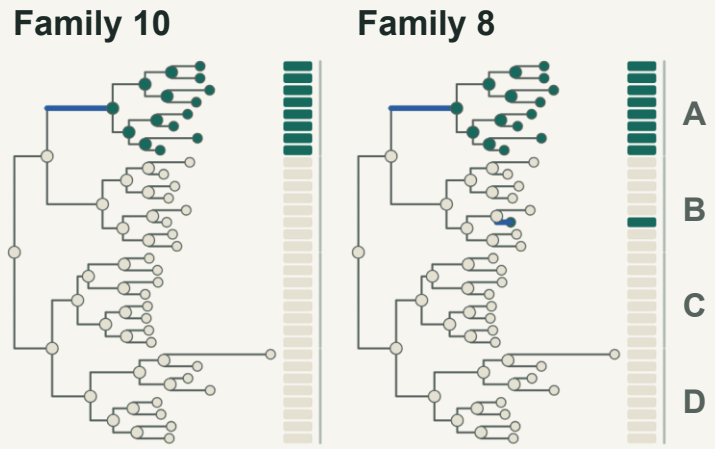
What we simulated: modules and decoys

Planted module M1



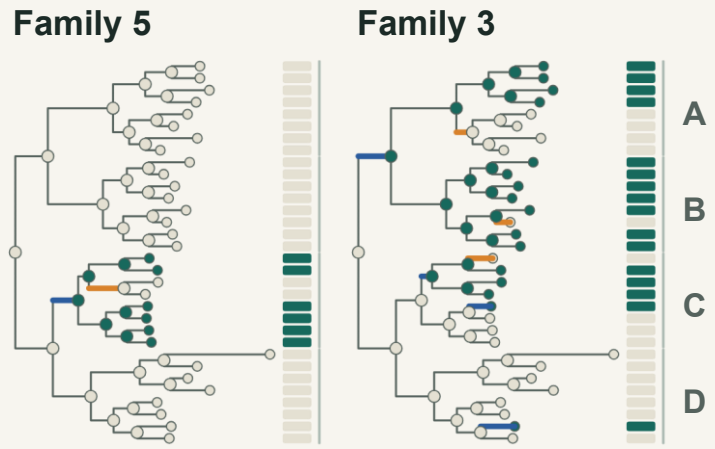
Gained together in the same 6 lineages: real co-evolution.

Clade decoys



Both live in clade A: they match through shared ancestry only.

Background



Gains in different places: unrelated.

32 genomes in four clades of eight, with unequal branch lengths (one long branch: D1), and 60 families: 5 + 4 module, 14 decoy and 37 background. All simulated, with invented annotation, for teaching only.

Fitted by GLOOME: pi1 = 0.45 (share of genomes carrying a family), gamma shape alpha = 3.0, four rate categories.

Source: simulated tree and phyletic pattern; GLOOME TreeAttributes.tsv. Blue = gain, orange = loss (parsimony).

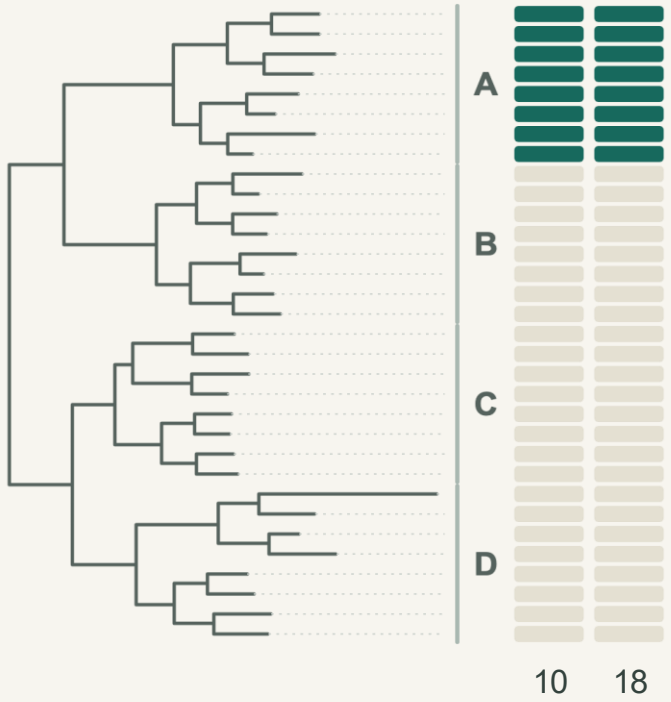
Relatedness can fake a correlation

Families 10 and 18 are both present in exactly the eight genomes of clade A, and nowhere else.

Their presence patterns are identical: r = 1.00.

But that is one shared ancestral gain, not two events that happened together: 8 genomes, one piece of evidence.

We must account for this. CoPAP does it by simulating unrelated positions on the same tree with the same model parameters: the null of what the tree alone produces. Here: p = 0.011, q = 0.43, not significant.



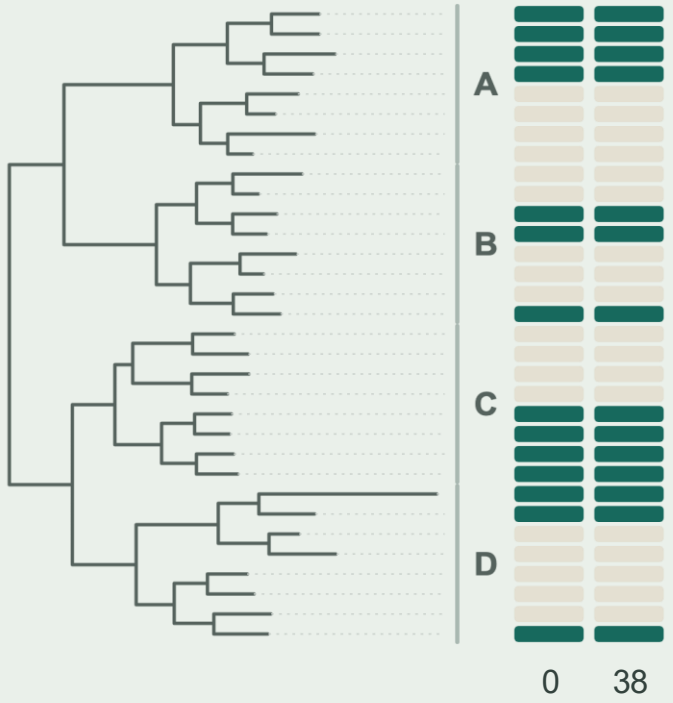
Source: simulated data, families 10 and 18; Coevolution.tsv

Independent repeats are the real evidence

Families 0 and 38 are present in the same 14 genomes, in six small groups spread over all four clades.

Each group is a separate gain of both families together.

Six coincidences are hard to explain by chance: r = 1.00, q = 0.028.



Source: simulated data, families 0 and 38; Coevolution.tsv

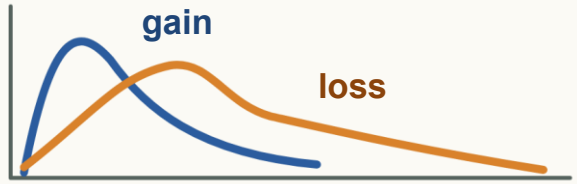
How CoPAP corrects for phylogeny

1 Input



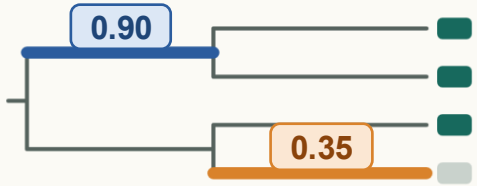
Presence and absence of each family, plus the species tree.

2 Model



GLOOME fits gain and loss rates on the tree, family by family.

3 Stochastic mapping



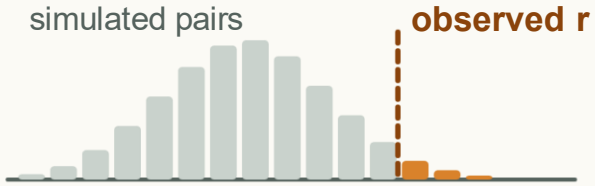
A gain or loss probability on every branch: a clade-wide gene is one event.

4 Correlation



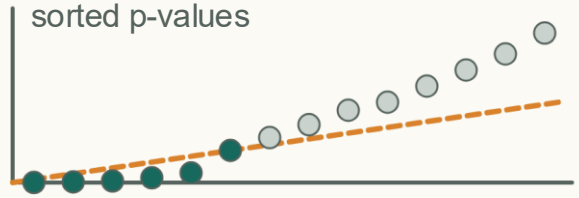
Each family is a vector over 4n−4 branch positions. Pearson r compares two vectors.

5 Simulations



Null pairs are simulated on this tree, matched by speed. p = share with r at least as high.

6 Multiple testing



1,770 pairs are tested at once, so we control the false discovery rate (FDR).

Workflow after CoPAP: Cohen, Ashkenazy, Levy Karin, Burstein & Pupko 2013 NAR 41:W232–W237; method: Cohen et al. 2012 Bioinformatics 28:i389–i394. Graphics illustrative.

What the CoPAP paper shows

Precision-recall (AUPR) on simulated data, higher is better



Speed

CoPAP needed under 1% of BayesTraits run time, so thousands of families are feasible.

Scale

4,258 families in 681 genomes gave 5,605 significant pairs.

Meaning

Over 90% of the 54 largest clusters were functionally coherent.

Source: CoPAP paper (Cohen et al., 2013), simulation benchmark and COG case study; AUPR values as reported there

Toy results: modules recovered, decoys filtered

40

pairs with presence r of 0.7 or more; 25 involve clade decoys

18

pairs with q below 0.05; 15 are the planted module pairs

23 of 25

decoy pairs removed by the null model

Module	Families (POS)	What it is
M1	0, 32, 38, 40, 58	the planted toxin island (all 5 found, 10 of 10 pairs)
M2	16, 19, 27, 54	the planted effector module (all 4 found, 5 of 6 pairs)
no module	8–12, 17–45, 25–53	the three false pairs stay isolated and form no module

Source: Coevolution.tsv (1,770 pairs) from the toy run; modules from copap_modules.R at q < 0.05

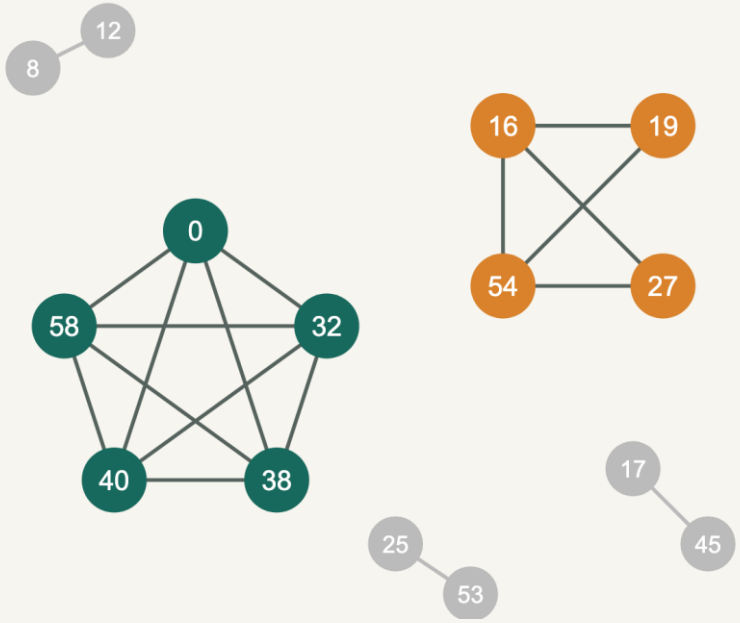
From pairs to modules

1. Keep the significant pairs (q below 0.05) and draw them as a network.

2. Transitivity clustering cuts the network into groups where nearly everyone is linked to everyone.

3. Read each module with its annotation: here a toxin island and an effector with its chaperone.

4. Isolated links stay out. The three false pairs (grey) form no module, but a module is still a hypothesis: check annotation and neighbourhood.



Source: simulated toy run, copap_modules.R at q < 0.05; method as in the CoPAP paper (Transitivity Clustering)

Gene-content GWAS: the same problem

CoPAP (previous section)

Does gene family **X** gain/loss track gene family **Y**?

Gene-content GWAS

Does gene family **X** gain/loss track a **phenotype** (host, niche, virulence, resistance)?

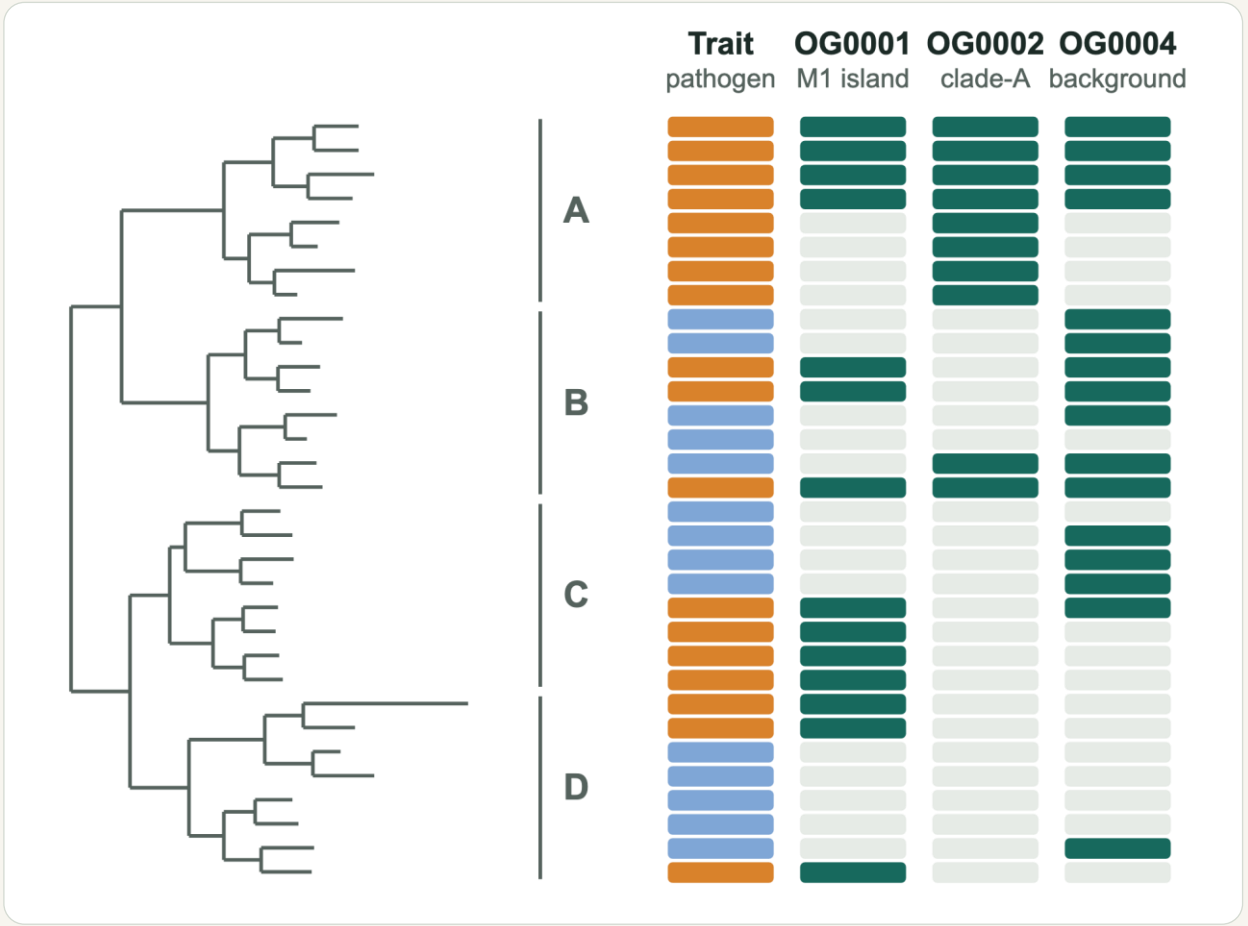
The same trap

Closely related genomes share both genes and traits by descent. A naive test (Fisher, chi-square) on all genomes finds hundreds of “associations” that only reflect the tree.

The same fix: **use the phylogeny**. The question changes from “how many genomes have both?” to “how many *independent evolutionary events* have both?”

Concept: Collins and Didelot (2018) PLoS Comput Biol 14:e1005958; Lees et al. (2018) Bioinformatics 34:4310

A toy example: pathogens and commensals



Fake trait, real tree

- Same 32 genomes and tree as before; **pathogen** or **commensal**
- Clade A is pathogenic by descent
- In B–D, pathogen only with the **M1 island**

Three gene families

- **OG0001**: M1, truly linked
- **OG0002**: only in clade A, tracks the trait by descent
- **OG0004**: unrelated background

Simulated data: same toy tree as the GLOOME/CoPAP example, fake trait

A naive test finds more than the truth

Naive test, one gene at a time

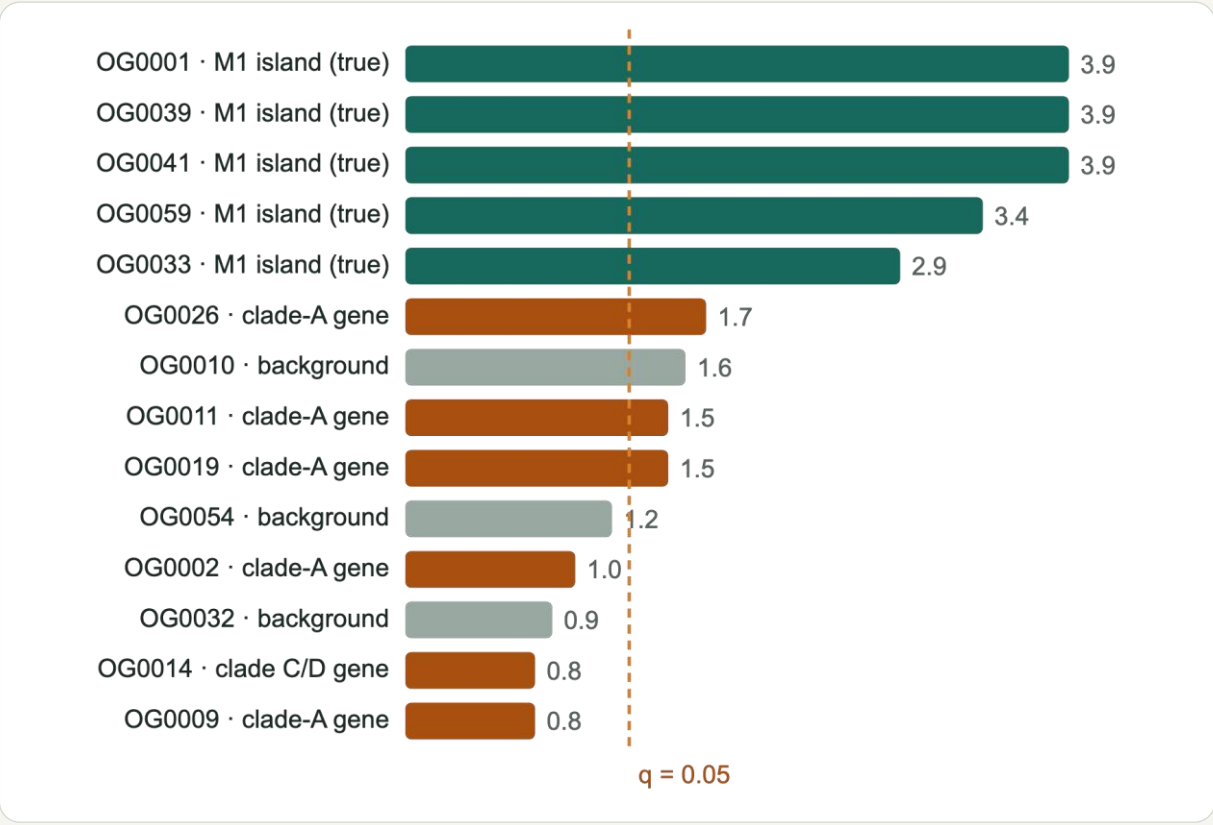
Null: the gene and the trait are independent across the 32 genomes. Fisher's exact test on this table:

OG0011	pathogen	commensal
present	8	0
absent	10	14

Clade-A gene: p = 0.0044. All 60 p-values are then corrected for multiple testing (FDR, Benjamini–Hochberg): q = 0.030.

Nine hits at q < 0.05

5 M1 genes (true), plus **3** clade-A genes and **1** background gene (false). FDR corrects for the number of tests, not for genomes that are not independent: clade genomes share genes and trait by descent.



Bar length = −log10(q), where q is the FDR-adjusted p-value. Dashed line: q = 0.05.

Simulated data: same toy tree as the GLOOME/CoPAP example, fake trait

treeWAS: phylogeny-aware association

treeWAS: simulate the null on the tree

- Input: binary matrix (SNPs or **gene presence/absence**), tree, binary or continuous phenotype
- Reconstruct ancestral states of genotype and phenotype on the tree
- Simulate a **null** dataset on the same tree, matching the homoplasy of the real data
- Three scores per gene: **terminal** (correlation across tips), **simultaneous** (changes on the same branches), **subsequent** (agreement of states along branches)
- Significant: beyond the null distribution of the same score

Same idea as CoPAP

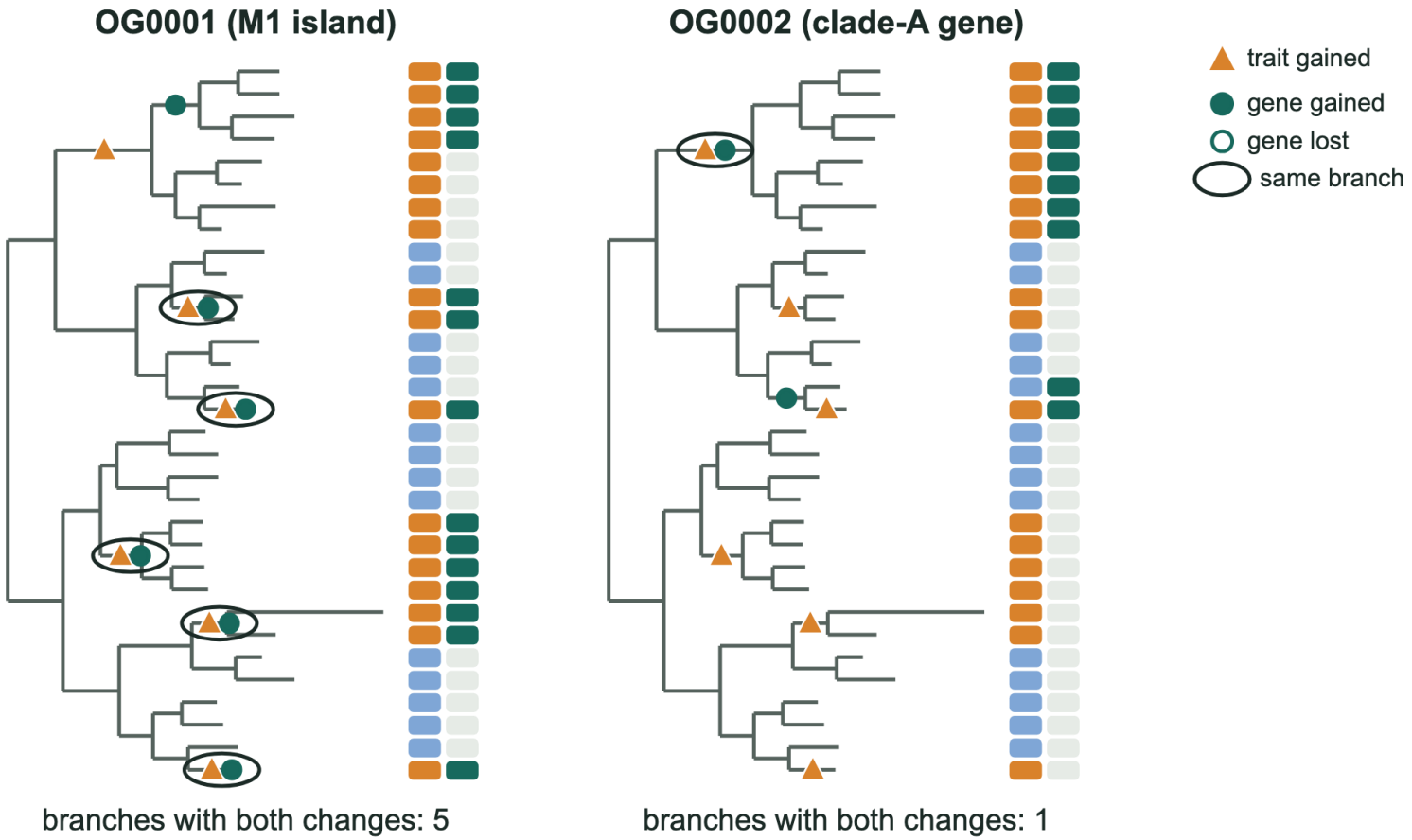
- Count events along branches, not genomes
- Compare with a simulated null on the same tree
- Differences: CoPAP is a model-based gain/loss analysis between genes and corrects for rate heterogeneity; treeWAS defaults to parsimony reconstruction (ML optional) and tests against a phenotype

R package, simple input. Simulating a null for each family is heavy for tens of thousands of families: filter first.

Source: Collins and Didelot (2018) PLoS Comput Biol 14:e1005958; treeWAS documentation (github.com/caitiecollins/treeWAS)

SUPPLEMENTARY · skip if short on time

treeWAS: events on the same branch



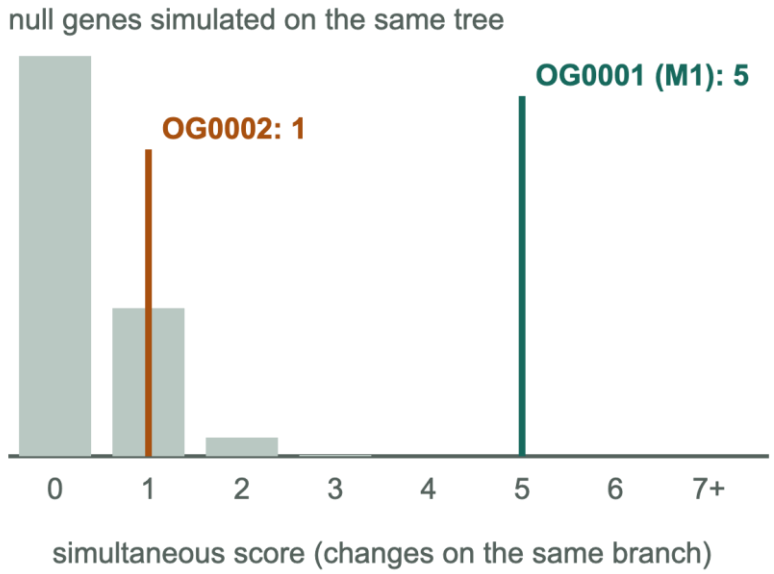
5 versus 1

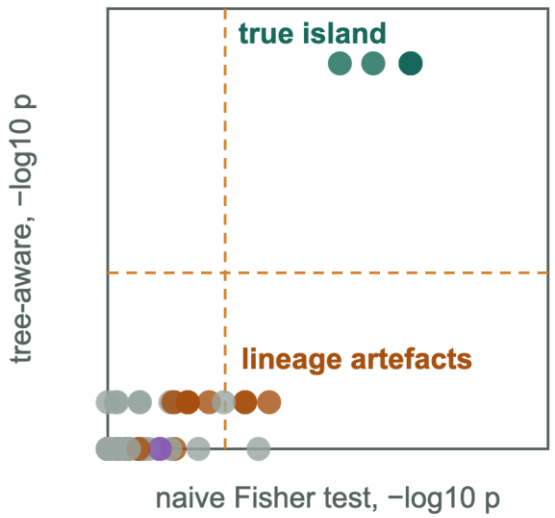
M1: gene and trait change together on **5** separate branches. Clade-A gene: **once**, at the base of the clade.

The tree turns “how many genomes share both?” into “how many **independent events** share both?”. We compare these scores with chance on the next slides.

Simulated data: same toy tree as the GLOOME/CoPAP example, fake trait; illustrative re-implementation of the simultaneous score

Is 5 more than expected?





Null: genes simulated on the same tree with the same number of changes. Score 5 never occurs in about 24,000 null genes; score 1 is common.

All 60 families: the 4 lineage artefacts are significant naively but not tree-aware. The 5 M1 genes stay.

Simulated data: same toy tree as the GLOOME/CoPAP example, fake trait; ~24,000 simulated null genes; dashed lines: p = 0.01

SUPPLEMENTARY · skip if short on time

Genes or k-mers as variants?

	Gene presence/absence	Unitigs and k-mers
Needs	Annotation and a pangenome	Assemblies only
Sees	Coding gene content	Any sequence: genes, SNPs, intergenic and regulatory regions
Result	A gene family, with function	A sequence, to be mapped back to genes or regions
Speed	Fast once the matrix exists	Many more variants to test

Gene presence/absence is easy to interpret but blind to **intergenic** and allelic variation. K-mers see it, at the price of **interpretability**. pyseer can test both; treeWAS takes any binary matrix.

Sources: Lees et al. (2018) Bioinformatics 34:4310; Collins and Didelot (2018) PLoS Comput Biol 14:e1005958

pyseer: linear mixed models

pyseer: regression with population structure

- Linear models with **fixed** (distance MDS) or **mixed** effects (kinship from the tree)
- Variants: **gene presence/absence** (Roary/Panaroo-style table), unitigs, k-mers, SNPs
- Binary or continuous traits; effect sizes with confidence intervals
- Lineage effects and heritability estimates

```
# kinship matrix from the species tree
phylogeny_distance.py --lmm tree.nwk \
> phylogeny_K.tsv

# gene P/A association
pyseer --lmm --phenotypes pheno.tsv \
--pres gene_presence_absence.Rtab \
--similarity phylogeny_K.tsv > gwas.txt
```

Source: Lees et al. (2018) Bioinformatics 34:4310; pyseer.readthedocs.io; check option names for your version

SUPPLEMENTARY · skip if short on time

treeWAS or pyseer?

	treeWAS	pyseer
Idea	Events on the tree against a simulated null	Regression with a relatedness correction
Genotype	Binary: SNPs or gene P/A	Gene P/A, unitigs, k-mers, SNPs
Strength	Same logic as CoPAP	Scales; standard, well documented
Watch out	Simulation cost; needs a good tree	Structure correction; independent origins

Rule of thumb

Just learnt CoPAP, with a table of gene presence and absence? **treeWAS** is the natural next step. Larger dataset, or k-mers and unitigs? **pyseer**.

Sources: Collins and Didelot (2018); Lees et al. (2018). Both limited by the number of independent origins of the trait

Reading gains and losses with care

A gain is not always HGT

Missing annotation, assembly gaps and orthogroup splits also look like gains and losses.

The tree matters

Events are placed on the tree you provide. A wrong tree or root changes the answer.

Neighbouring branches are ambiguous

Family 5: a gain on N18 (0.45), N22 (0.43) or N20 (0.42) are almost equally likely.

Decoys can slip through

In the toy data, 2 of 25 decoy pairs still passed. They stayed isolated, but they can mislead.

Source: BranchPositionProbabilities.tsv and Coevolution.tsv

Hands-on session after the break: Pseudomonas dataset

Microbializer
orthogroups, phyletic pattern, core tree

→

GLOOME
homoplasy, rates, gain and loss probabilities

→

CoPAP / GOLDFINDER
families that move together

→

Interpretation
link events to biology and to the host

Questions to bring to your own data

- Which families are scattered (low CI), and which follow the tree?
- On which branches do gains and losses cluster?
- Which families gain and lose together?

Materials, handout and data: github.com/HaimAshk/bacterial-phylogenomics-workshop

Same steps as the toy example, applied to the Pseudomonas genome set